

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO2	Assessment	Scenario-Based
Weightage		Total Marks	

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Section/Batch	AIDS-2023	Date of Submission	
Faculty Incharge	DR.R S Selvan	Assessment	Individual Submission

TOTAL MARKS	20	OBTAINED MARKS	
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COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I V.Sree Ruthin Reddy (192324112) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V. Sree Ruthin Reddy

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

Scenario :

A smart campus has been allocated the IP block 192.168.100.0/24. Different departments require different Subnet sizes.

DepartmentSubnetmask

School of Computing 125

School of Electronics 126

School of mechanical 127.

Engineering

Subnet Allocation.

1) School of Computing (125)

Subnet mask

255.255.255.128.

Block size

128

Network Address

192.168.100.0

Broadcast Address. 192.168.100.127.

Valid host Range

192.168.100.1 - 192.168.100.126.

usable hosts : $2^{(32-25)} - 2$

= 126.

2) School of Electronics

Subnet mask : 255 : 255 : 255 : 192

Block size : 64

Network address = 192.168.100.191

Broadcast Address : 192.168.100.191

Usable hosts : $2^6 - 2 = 62$

3) School of Mechanical Engineering (124)

Subnet mask : 255 : 255 : 255 : 224

Block size : 32

Network address : 192.168.100.192

Usable hosts : $2^5 - 2 = 30$

2)

Scenario -

Company network, 10.10.0.0/16.

1) Software Development (124)

Subnet mask : 255.255.255.0

Network address : 10.10.0.0

Broadcast Address : 10.10.0.255

Host Range : 10.10.0.1 - 10.10.0.254

Usable hosts : $2^8 - 2 = 254$

2) Quality Assurance (125)

Network Address = 10.10.1.10

Broadcast : 10.10.1.129

Host Range : 10.10.1.1 - 10.10.1.128

3) Human Resources (126)

Network Address : $10.10.1.128$

Broadcast Address : $10.10.1.191$

Host Range : $10.10.1.129 - 10.10.1.190$

Usable Hosts : 62.

4) Executive Management (127)

Network Address : $10.10.1.192$

Allocated : $10.10.0.0 - 10.10.1.223$

Unused : $10.10.1.224 - 10.10.255.255$

IPv6 address.

$2001:0db8:abcd:0000:0000:0000:0000:0000/64$

1) Compressed form

Leading zeros removed

Consecutive zero blocks replaced by :

$\rightarrow 2001:db8:abcd::1/64$

2) Expanded form

$2001:0db8:abcd:0000:0000:0000:0000:0000$

3) Network prefix

Prefix Length : 64

Network prefix : $2001:db8:abcd:0000::1/64$

Host portion : Last 64 bits.

4) Number of Host Addresses.

Remaining Bits : $128 - 64 = 64$

4)

Routing Table.

Destination Network

192.168.1.0/24

192.168.2.0/24

Destination IP

192.168.2.45

1) Destination IP

192.168.2.45.

2) matching Routing Entry.

192.168.2.0/24.

3) Next Hop

Packet is forwarded

through the interface connected to

192.168.2.0/24 network.

4) Default Route

If no Entry matches Defaults Route: 0.0.0.0/0

The Router forwards the packet to the configured default gateway.

5)

Administration LAN

192.168.10.0/26.

Finance LAN: 192.168.10.64/26.

Gateways Administration: 192.168.10.1

Finance: 192.168.10.64

Administration LAN.

Network: 192.168.10.0

Broadcast: 192.168.10.63.

Host Range: 192.168.10.1 - 192.168.10.62

Gateway: 192.168.10.65